

3 Partial Derivatives

Example 3.4.1 The base radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.01 cm in each. Use a tangent plane to estimate the maximum error in the calculated volume of the cone.

3.5 The Chain Rule

Theorem 3.5.1 (*The Chain Rule Version 1*) Let $f(x, y)$ be a differentiable function, and $x = g(t)$ and $y = h(t)$ are both differentiable functions. Then

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Example 3.5.2 If $z = x^2y + 3xy^4$ where $x = \cos t$ and $y = \sin 3t$, find $\frac{dz}{dt}$ at $t = 0$.

Theorem 3.5.3 (*The Chain Rule Version 2*) Let $f(x, y)$ be a differentiable function where $x = g(s, t)$ and $y = h(s, t)$ are both differentiable functions of two variables, then

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} \quad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}.$$

Example 3.5.4 If $z = e^x \sin y$ and $x = st^2$ and $y = s^2t$, find both $\partial z/\partial s$ and $\partial z/\partial t$.

Example 3.5.5 If $z = f(x, y)$ has continuous second-order partial derivatives, and $x = s^2 + t^2$ and $y = 2st$, find expressions for $\partial z/\partial s$ and $\partial^2 z/\partial^2 s$.

Example 3.5.6 Let $f(x, y) = x \sin(y)$ and suppose that $x = 2st$ and $y = s^3$. Find $\partial f/\partial s$ and $\partial f/\partial t$.

3.5.1 Implicit Differentiation

We can use the chain rule to obtain an application to implicit differentiation from Calculus I:

Example 3.5.7 Find y' if $x^3 + y^3 = 6xy$.